



# Nguyen Ngoc Hoang Duong

## Software Developer

|               |                                               |
|---------------|-----------------------------------------------|
| Date of birth | June 13, 2005                                 |
| Gender        | Male                                          |
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## SUMMARY

Careful, responsible, and detail-oriented Software Engineering student. I prefer to understand a problem clearly before implementing it, keep my work organised, and verify important behaviour instead of assuming that it works. I am willing to learn unfamiliar technologies when a project requires them, while still paying attention to readable code, consistent interfaces, and complete results.

## TECHNICAL SKILLS

### Frontend

Vue 3, Vite, React, Flutter, Dart, TypeScript, JavaScript and responsive interface development

### Backend

Node.js, Express, REST APIs, Socket.IO, JWT, authentication and asynchronous processing

### Applied AI and Data

Python, custom RAG pipelines, nomic-embed-text embeddings, prompt construction, structured-output validation, Docling, OCR, evidence validation, evaluation and Ollama

### Databases

MongoDB, PostgreSQL, MySQL, Firebase, Room DB and SQLite

### Tools and Deployment

Git, GitHub, Docker, Postman, Render, MongoDB Atlas, Cloudinary and Firebase Storage

## EDUCATION

2023 - 2026

### University of Greenwich Vietnam

#### Computing - Software Engineering

- Current GPA: 3.5 / 4.0 (before the final project grade is included)
- Studied Data Structures and Algorithms, Software Engineering, Agile Development with Scrum, Advanced Microservices Development, and Data and Web Analytics.
- Completed and successfully defended DreamScape as the final-year project.

## WORK EXPERIENCE

Sep 2025 - Dec 2025

### Techtack

#### Frontend Developer Intern

- Joined the ESA Pay project during its early development and implemented Vue 3 interfaces for accounting and banking workflows.
- Worked with backend and testing tasks under frequent deadlines, while reporting progress and integration issues each day.

## SELECTED PROJECTS

Sep 2025 - Dec 2025

### ESA Pay

#### Frontend Developer Intern at Techtack

- **Business workflows:** implemented receivables, payables, accounting vouchers, invoices, and bank-related flows after studying product and API documentation.
- **Large API integration:** connected more than 100 REST endpoints to Vue screens and handled data mapping, validation, loading, empty, and error states.
- **Banking tests and data alignment:** tested banking APIs, compared payloads with provider documents, identified mismatches between frontend and backend structures, and proposed field or schema adjustments.
- **Delivery discipline:** met frequent deadlines, reported progress daily, and raised integration blockers early during the project's initial development.

*The source code and project materials remain private company property.*

Feb 2026 - Apr 2026

### Expense Tracker System

#### Mobile and Full-stack Developer

- **Offline-first user application:** used Flutter and Dart for one mobile codebase, while Room DB and SQLite kept expense entry and history available without an internet connection.
- **Native administration application:** used Android and Java to implement account management, monitoring, and financial analytics while practising platform-specific application design.
- **Realtime synchronisation:** connected Firebase Authentication, Firestore, and Realtime Database through a custom queue so local changes were preserved offline and synchronised when connectivity returned.
- **HCI decisions:** kept transaction actions predictable, used clear form validation and immediate feedback, and separated user and administrator workflows so each interface displayed only the controls needed for its role.

**User App** : [github.com/HelloDuongNha/expense-tracker-user](https://github.com/HelloDuongNha/expense-tracker-user) **Admin App** : [github.com/HelloDuongNha/expense-tracker-admin](https://github.com/HelloDuongNha/expense-tracker-admin)

Apr 2026 - Aug 2026

DreamScape

Project Owner, Full-stack Developer and AI/RAG Pipeline Developer

- **Responsive and HCI design:** built one EN-VI Vue 3 and TypeScript interface for desktop and mobile, with consistent navigation, visible processing states, confirmations, and citations placed near AI-supported content.
- **Modular backend:** separated Express and Mongoose code into authentication, dream, community, messaging, academic knowledge, and Oracle modules without the operational cost of microservices.
- **Automated source pipeline:** resolved DOI metadata and HTML or XML full text, then used PDF, Docling, and OCR fallbacks. Cleaned sections were chunked and embedded before administrator review.
- **Rule and evidence automation:** extracted candidate rules after approval, linked exact evidence passages, matched new sources to unresolved evidence needs, and kept unsupported claims as visible gaps.
- **Grounded AI analysis:** used nomic-embed-text and custom RAG retrieval to select reviewed rules, evidence, personal history, and permitted similar dreams before Oracle generation; backend checks validated structure and citations.
- **Realtime infrastructure and deployment:** combined Socket.IO, MongoDB Atlas, Redis, Cloudinary, and Firebase Storage; deployed the Vue site and Express API on Render, with Ollama, embeddings, Docling, and OCR on a remote NVIDIA T4 environment.

 **Live Product** : [dreamscape-fe.onrender.com](https://dreamscape-fe.onrender.com)

 **Frontend** : [github.com/HelloDuongNha/Dreamscape-FE](https://github.com/HelloDuongNha/Dreamscape-FE)

 **Backend** : [github.com/HelloDuongNha/Dreamscape-BE](https://github.com/HelloDuongNha/Dreamscape-BE)

ACTIVITIES

May 2025 - Apr 2026

Greenwich IT Club

Head of External Relations

Built relationships with companies and universities, managed outreach, and co-organised technology events for the student community.

CERTIFICATIONS

2022  
IELTS 5.5

HONORS AND AWARDS

Top 3 students of trimester  
Spring 2025 and Summer 2025

Best student by subject  
Mathematics for Computer, Object-Oriented Programming, User Interface Design, and Application Development

INTERESTS

- **Sports:** badminton, gym training, and swimming.
- **Technology and hardware:** custom PC building, hardware optimisation, audio equipment, and wearable technology.
- **Design and craftsmanship:** structural design, material quality, and practical product aesthetics.